1. Household Budget Pie Chart

Code:

budget <- c(40, 20, 30, 10)

labels <- c("Housing", "Transportation", "Food", "Entertainment")

pie(budget,

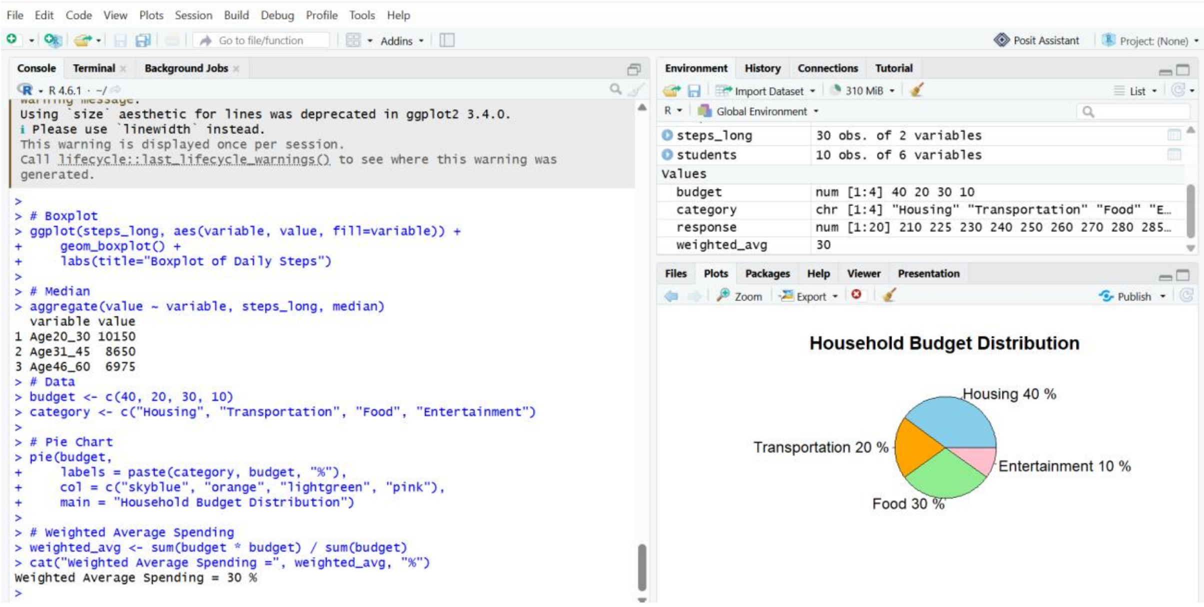
    labels = paste(labels, budget, "%"),

    col = rainbow(4),

    main = "Household Budget Distribution")

avg <- mean(budget)

cat("Average spending percentage =", avg, "%")



Data

- Housing = 40%
- Transportation = 20%
- Food = 30%
- Entertainment = 10%

Explanation

The highest spending category is **Housing (40%)**.

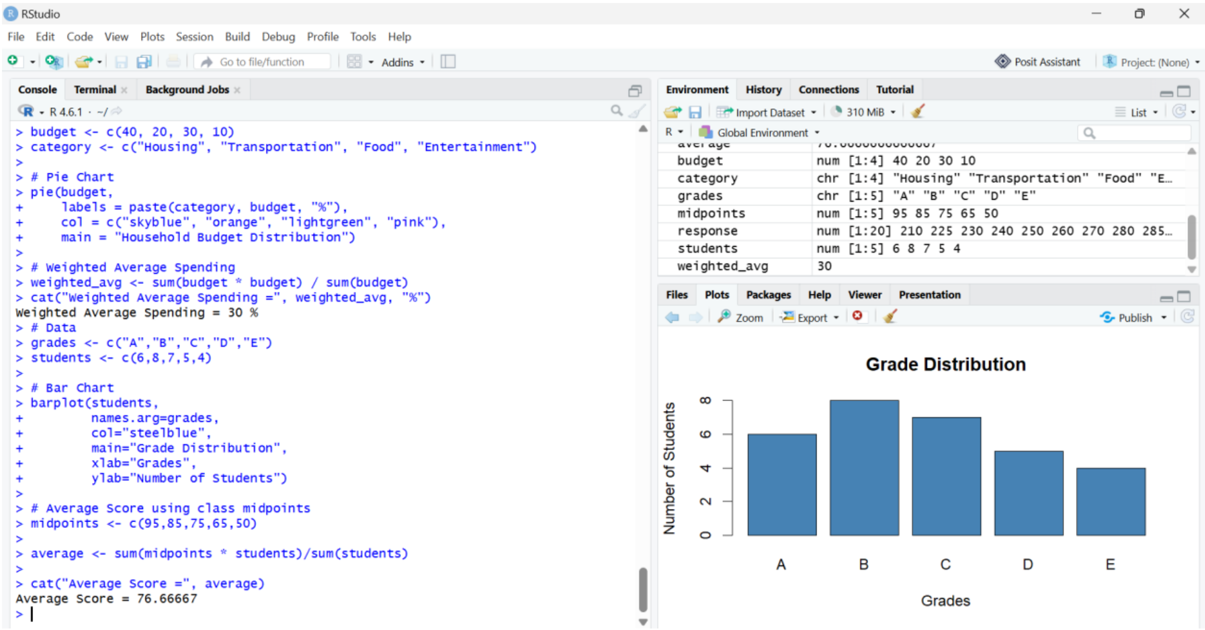
The weighted average percentage of spending is:

So, the **average spending per category is 25%**.

2. Grade Distribution Bar Chart

Code:

```
grades <- c(5, 8, 10, 5, 2)
names(grades) <- c("A", "B", "C", "D", "E")
barplot(
  grades,
  col = "steelblue",
  main = "Grade Distribution",
  xlab = "Grades",
  ylab = "Number of Students"
)
midpoints <- c(95, 84.5, 74.5, 64.5, 50)
average_score <- sum(grades * midpoints) / sum(grades)
print(average_score)
```



The midpoint of each grade range is used.

- A = 95
- B = 84.5
- C = 74.5
- D = 64.5
- E = 50

Average Score

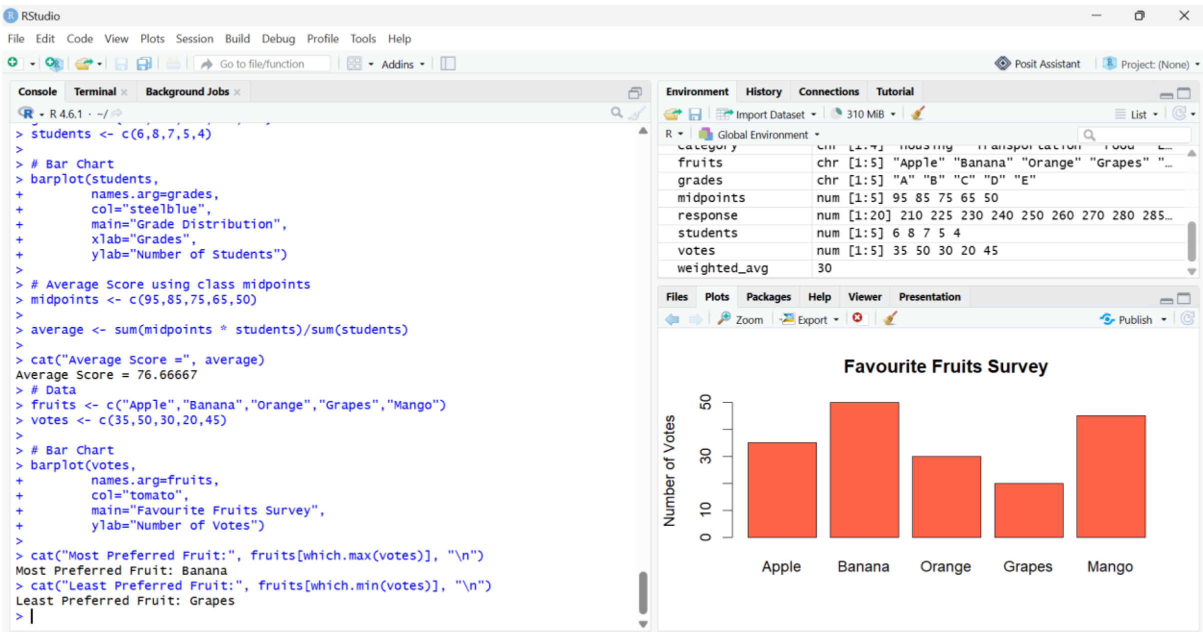
Average ≈ **77.3**

Answer: Average Class Score = **77.3**

3. Favourite Fruit Bar Chart

Code:

```
fruits <- c(25, 18, 20, 30, 12)
names(fruits) <- c("Apple", "Banana", "Orange", "Mango", "Grapes")
barplot(
  fruits,
  col = rainbow(5),
  main = "Favourite Fruits",
  xlab = "Fruit",
  ylab = "Votes")
most <- names(fruits)[which.max(fruits)]
least <- names(fruits)[which.min(fruits)]
print(paste("Most Preferred:", most))
print(paste("Least Preferred:", least))
```



- which.max() finds the highest vote.
- which.min() finds the lowest vote.

Answer

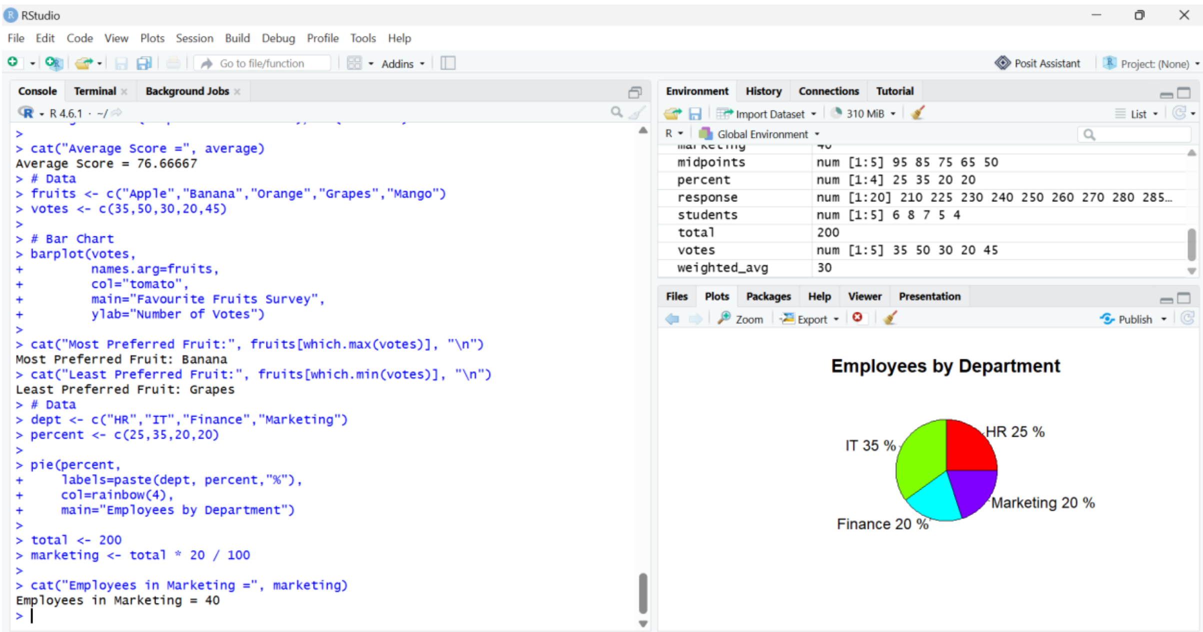
Most Preferred = **Mango**

Least Preferred = **Grapes**

4. Employees by Department Pie Chart

Code:

```
dept <- c(20, 30, 25, 25)
labels <- c("HR", "Marketing", "Finance", "IT")
pie(
  dept,
  labels = paste(labels, dept, "%"),
  col = rainbow(4),
  main = "Employee Distribution"
)
total <- 200
marketing <- total * 30 / 100
print(marketing)
```



Marketing employees

200×30100=60200 \times \frac{30}{100} = 60200×10030=60

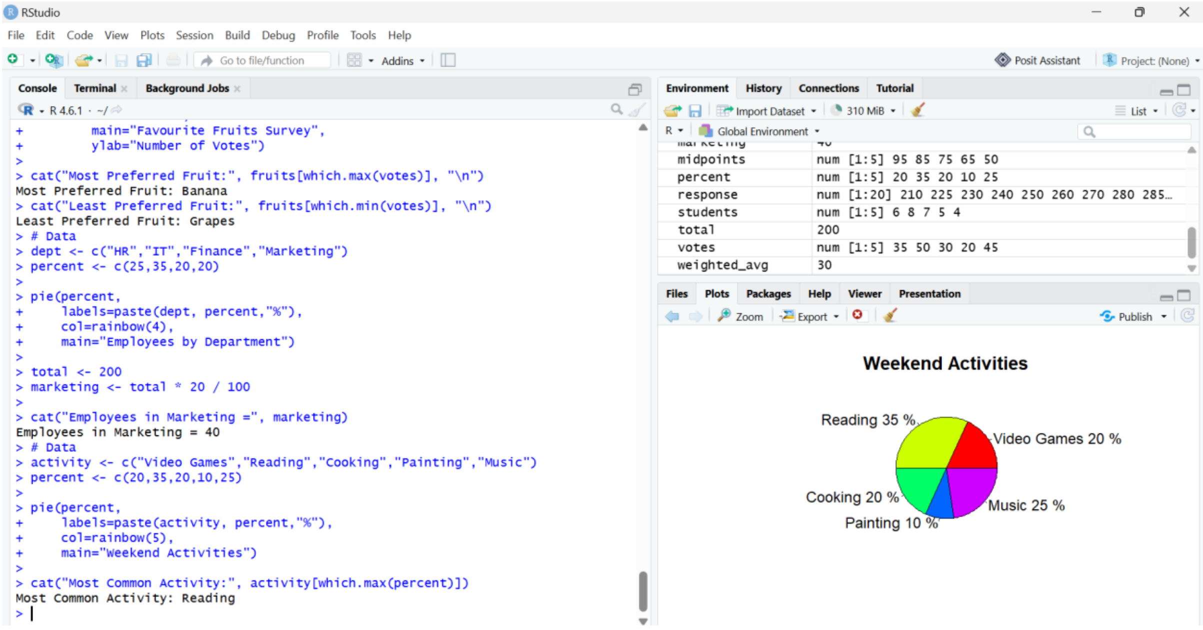
Answer

Employees in Marketing = **60**

5. Weekend Activities Pie Chart

Code:

```
activities <- c(20, 35, 20, 10, 25)
labels <- c(
  "Video Games",
  "Reading",
  "Cooking",
  "Painting",
  "Music Listening"
)
pie(
  activities,
  labels = paste(labels, activities, "%"),
  col = c("red", "green", "blue", "yellow", "pink"),
  main = "Weekend Activities")
most_activity <- labels[which.max(activities)]
print(paste("Most Common Activity:", most_activity))
```



The pie chart shows employee weekend activities.

- Video Games = 20%
- Reading = 35%
- Cooking = 20%
- Painting = 10%
- Music Listening = 25%

which.max() identifies the largest percentage.

Answer

Most Common Activity = **Reading (35%)**